

Institut universitaire des sciences
Faculté des sciences et des technologies

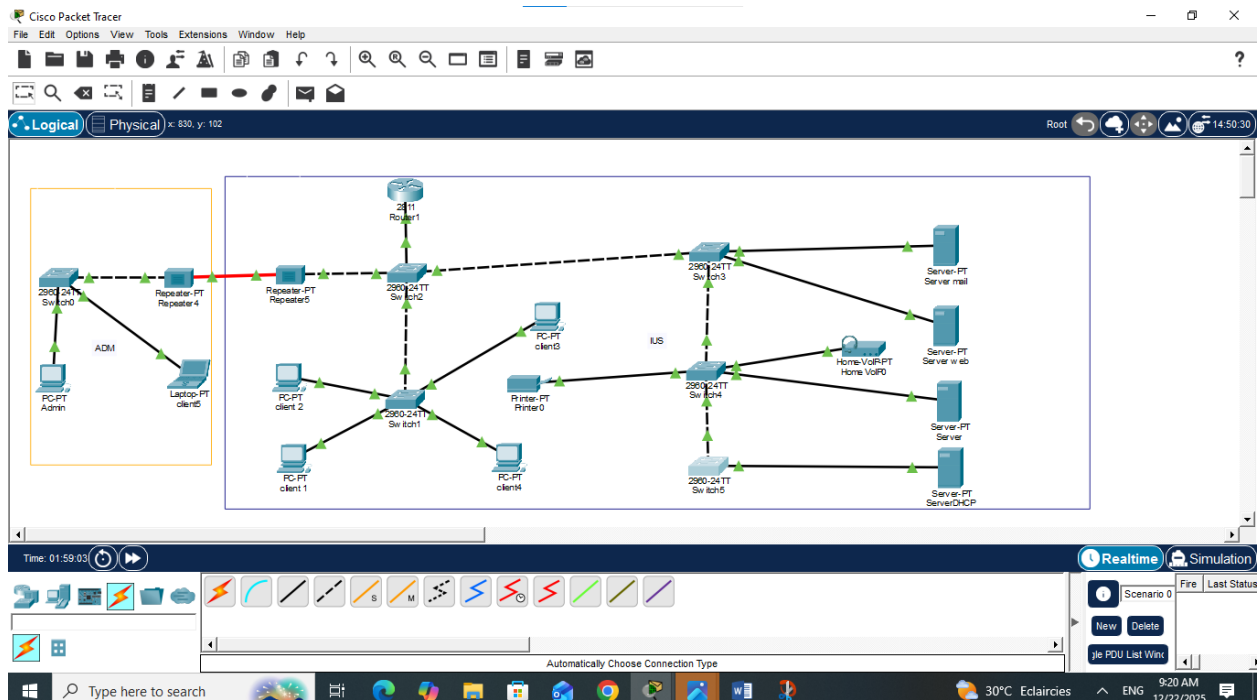
TD N°6 – Réseaux I

Nom & Prénom : Leamande Evena

Niveau: L3/fst

Date: 22/12/25

Objectif: L'idée de ce TD, c'était d'apprendre à ne plus configurer chaque ordinateur un par un à la main. J'ai configuré le routeur pour qu'il devienne le 'cerveau' de la distribution des adresses. Mon but était que, dès qu'un appareil se branche sur le réseau ADM ou IUS, il reçoive automatiquement son adresse IP (en IPv4 et en IPv6), sa passerelle pour aller sur internet et l'adresse du serveur DNS pour naviguer.



J'ai commencé par monter le réseau, Il y a deux grandes parties : le côté administration (ADM) et le côté technique(IUS). J'ai relié tout ça au routeur central qui fait le lien entre les deux mondes.

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip helper-address 192.168.1.1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ipv6 unicast-routing
Router(config)#int f0/0
Router(config-if)#ipv6 address 2001:DB8:1::1/64
Router(config-if)#ipv6 enable
Router(config-if)#no sh
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#int f0/1
Router(config-if)#ipv6 address 2001:DB8:2::1/64
Router(config-if)#ipv6 enable
Router(config-if)#no shutdown
```

Copy Paste

Top

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down

Router(config-if)#exit
Router(config)#ipv6 dhcp pool IUS
Router(config-dhcpv6)#address prefix 2001:DB8:1::/64
Router(config-dhcpv6)#dns-server 2001:4860:4860::8888
Router(config-dhcpv6)#domain-name reseau-IUS.com
Router(config-dhcpv6)#exit
Router(config)#ipv6 dhcp pool ADM
Router(config-dhcpv6)#address prefix 2001:DB8:2::/64
Router(config-dhcpv6)#dns-server 2001:4860:4860::8888
Router(config-dhcpv6)#domain-name reseau-ADM.com
Router(config-dhcpv6)#exit
Router(config)#int f0/0
Router(config-if)#ipv6 dhcp server IUS
Router(config-if)#ipv6 nd managed-config-flag
Router(config-if)#exit
Router(config)#int f0/1
Router(config-if)#ipv6 dhcp server ADM
Router(config-if)#ipv6 nd managed-config-flag
Router(config-if)#
Router(config-if)#exit
Router(config)#

Router con0 is now available

Press RETURN to get started
```

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Admin

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

192.168.1.3

Subnet Mask

255.255.255.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address

FE80::2E0:F9FF:FEA9:1841

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top



Type here to search



client 1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

192.168.1.6

Subnet Mask

255.255.255.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

☒ Automatic

☐ Static

IPv6 Address

2001:DB8:1:0:1326:EBFE:4F19:C068

Link Local Address

FE80::250:FFF:FE80:8122

Default Gateway

DNS Server

2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top



Type here to search



client 2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 192.168.1.4

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static

IPv6 Address: 2001:DB8:1:0:93D2:5C18:A3A1:165E

Link Local Address: FE80::2D0:97FF:FE12:ED7E

Default Gateway:

DNS Server: 2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Windows taskbar: Type here to search, Edge, File Explorer, Task View, Settings, Mail, Chrome, etc.

Configuration pour les

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP☐ Static

IPv4 Address

192.168.1.7

Subnet Mask

255.255.255.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

☒ Automatic☐ Static

IPv6 Address

2001:DB8:1:0:819:AA7:5130:EF11

Link Local Address

FE80::201:96FF:FE13:DD94

Default Gateway

DNS Server

2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

Type here to search



Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP☐ Static

IPv4 Address

192.168.1.8

Subnet Mask

255.255.255.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

☒ Automatic☐ Static

IPv6 Address

2001:DB8:1:0:70F8:48C0:9F59:1FA8

Link Local Address

FE80::2E0:A3FF:FEC3:3A35

Default Gateway

DNS Server

2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

Type here to search



Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP☐ Static

IPv4 Address

192.168.1.5

Subnet Mask

255.255.255.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

☒ Automatic☐ Static

IPv6 Address

2001:DB8:1:0:8A76:99A6:E02F:617E

Link Local Address

FE80::201:C7FF:FE91:4346

Default Gateway

DNS Server

2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

Type here to search



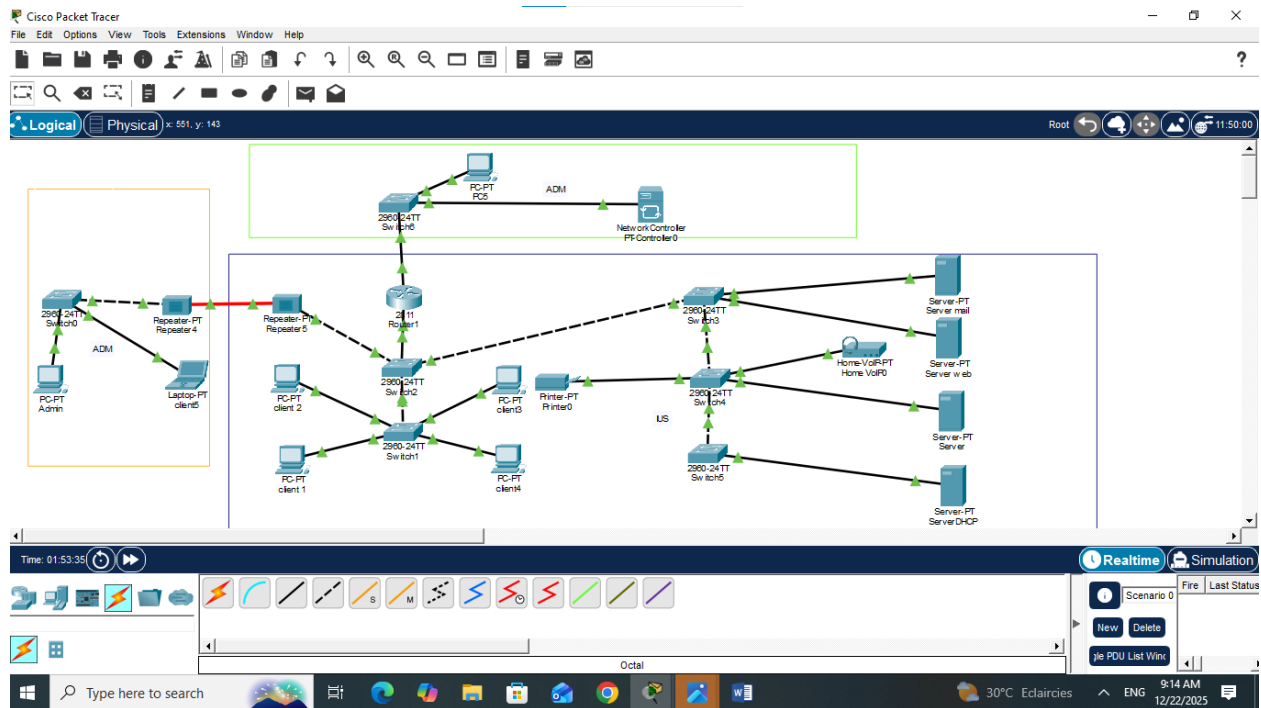
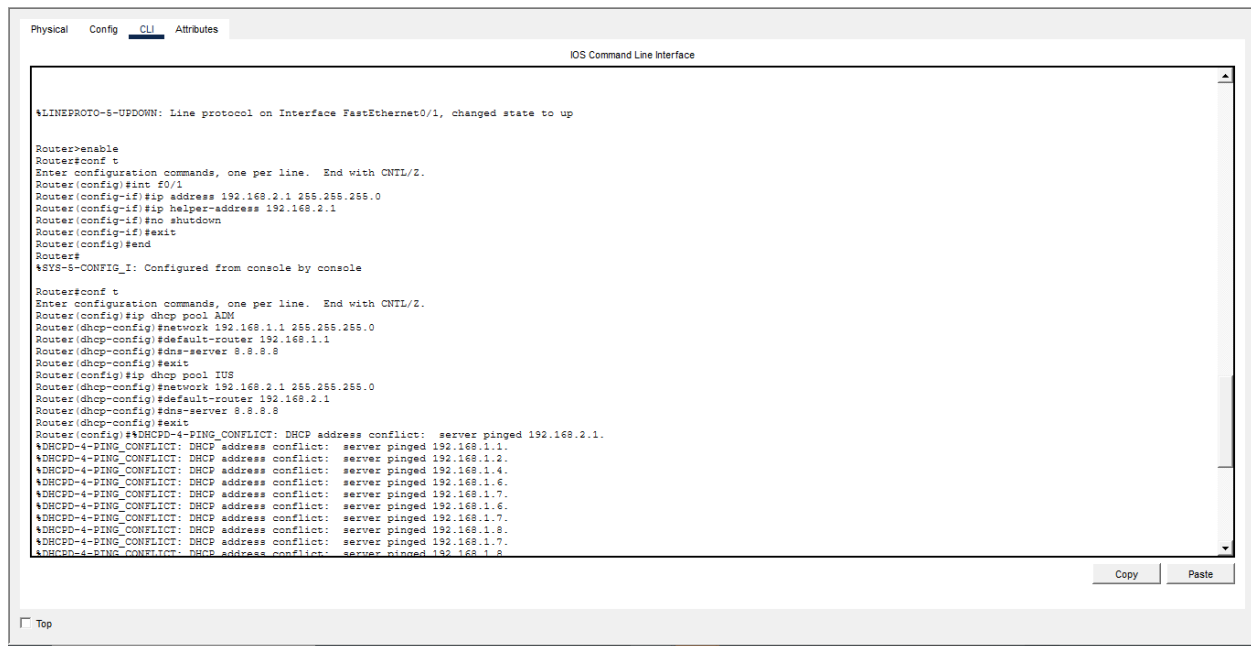


Figure 2



Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP☐ Static

IPv4 Address

192.168.2.2

Subnet Mask

255.255.255.0

Default Gateway

192.168.2.1

DNS Server

8.8.8.8

IPv6 Configuration

☒ Automatic☐ Static

IPv6 Address

2001:DB8:2:0:5541:8095:E5B0:660F

Link Local Address

FE80::201:C9FF:FE74:6406

Default Gateway

FE80::201:63FF:FE74:8C02

DNS Server

2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

Type here to search



Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP☐ Static

IPv4 Address

192.168.1.3

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

8.8.8.8

IPv6 Configuration

☒ Automatic☐ Static

IPv6 Address

2001:DB8:1:0:2A03:3934:723A:F37A

Link Local Address

FE80::2E0:F9FF:FEA9:1841

Default Gateway

FE80::201:63FF:FE74:8C01

DNS Server

2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

Type here to search



IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

192.168.1.6

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

8.8.8.8

IPv6 Configuration

☒ Automatic

☐ Static

IPv6 Address

2001:DB8:1:0:1326:EBFE:4F19:C068

Link Local Address

FE80::250:FFF:FE80:8122

Default Gateway

FE80::201:63FF:FE74:8C01

DNS Server

2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top



Type here to search



IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

192.168.1.4

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

8.8.8.8

IPv6 Configuration

☒ Automatic

☐ Static

IPv6 Address

2001:DB8:1:0:93D2:5C18:A3A1:165E

Link Local Address

FE80::2D0:97FF:FE12:ED7E

Default Gateway

FE80::201:63FF:FE74:8C01

DNS Server

2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top



Type here to search



Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP☐ Static

IPv4 Address

192.168.1.10

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

8.8.8.8

IPv6 Configuration

☒ Automatic☐ Static

IPv6 Address

2001:DB8:1:0:819:AA7:5130:EF11

Link Local Address

FE80::201:96FF:FE13:DD94

Default Gateway

FE80::201:63FF:FE74:8C01

DNS Server

2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

Type here to search



Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP☐ Static

IPv4 Address

192.168.1.8

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

8.8.8.8

IPv6 Configuration

☒ Automatic☐ Static

IPv6 Address

2001:DB8:1:0:70F8:48C0:9F59:1FA8

Link Local Address

FE80::2E0:A3FF:FEC3:3A35

Default Gateway

FE80::201:63FF:FE74:8C01

DNS Server

2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

Type here to search



Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP☐ Static

IPv4 Address

192.168.1.5

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

8.8.8.8

IPv6 Configuration

☒ Automatic☐ Static

IPv6 Address

2001:DB8:1:0:8A76:99A6:E02F:617E

Link Local Address

FE80::201:C7FF:FE91:4346

Default Gateway

FE80::201:63FF:FE74:8C01

DNS Server

2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

Type here to search



En conclusion :Ce que j'ai retenu de ce TD, c'est que le DHCP est indispensable dès qu'un réseau s'agrandit. Sans ça, on perdrait trop de temps et on ferait des erreurs de frappe. J'ai aussi appris à gérer les deux protocoles en même temps (IPv4 et IPv6), pour que le réseau soit compatible avec tout.